F18CF – Linear Algebra

Tutorial Sheet 7: Inner Product Spaces

Getting started.

1. We work in $\mathbb{R}^n$ with its standard (Euclidean) inner product.

   (a) Compute the orthogonal projection of $(1,2,3)^T \in \mathbb{R}^3$ on the line spanned by the vector $(1,1,1)^T$.

   (b) Check that the family $\vec{v}_1 = (1,1,1)^T$, $\vec{v}_2 = (1,-1,0)$, $\vec{v}_3 = (1,1,-2)^T$ is an orthogonal basis of $\mathbb{R}^3$. Write down the vector $(1,2,3)^T$ as a linear combination of $\vec{v}_1, \vec{v}_2, \vec{v}_3$.

   (c) Check that the family $\vec{v}_1 = (1,1,-1,0)^T, \vec{v}_2 = (0,1,1,-1)^T, \vec{v}_3 = (1,0,1,1)^T$ form an orthogonal family of vectors of $\mathbb{R}^4$. Compute the orthogonal projection of $(1,1,1,1)^T$ onto $\mathrm{span}(\vec{v}_1, \vec{v}_2, \vec{v}_3)$.

2. We work in $\mathbb{R}^n$ with its standard (Euclidean) inner product.

   (a) Check that $\vec{v}_1 = (1,1,1)^T, \vec{v}_2 = (0,1,1)^T, \vec{v}_3 = (0,0,1)^T$ forms a basis of $\mathbb{R}^3$. Apply the Gram-Schmidt algorithm to obtain an orthogonal basis of $\mathbb{R}^3$ containing $\vec{v}_1$.

   (b) Check that $\vec{v}_1 = (0,2,1,0)^T, \vec{v}_2 = (1,-1,0,0)^T, \vec{v}_3 = (1,2,0,-1)^T, \vec{v}_4 = (1,0,0,1)^T$ forms a basis of $\mathbb{R}^4$. Apply the Gram-Schmidt algorithm to obtain an orthogonal basis of $\mathbb{R}^4$ containing $\vec{v}_1$.

3. Show that the following formulas define inner products:

   (a) On $\mathbb{R}^3$, $\langle \vec{x}, \vec{y} \rangle := x_1y_1 + 2x_2y_2 + 3x_3y_3$.

   (b) On $\mathcal{C}_{2\pi}(\mathbb{R})$, the space of continuous functions from $\mathbb{R}$ to $\mathbb{R}$ that are 2π-periodic, $\langle f, g \rangle := \int_0^{2\pi} f(t)g(t)\mathrm{d}t$.

Problems.

4. **Computing projections.**

   (a) Show that the set

   $$H := \{(x,y,z,t)^T \in \mathbb{R}^4 \mid x + y + z + t = 0\}$$

   is a subspace of $\mathbb{R}^4$, and find a basis of H.

   (b) Find an orthogonal basis of H.

   (c) Compute the orthogonal projections of the following vectors on H:

   $$(i)\ \vec{v} = (1,2,3,4)^T, \qquad (ii)\ \vec{v} = (1,0,0,1)^T.$$

5. **A least squares problem.**

   (a) Show that there is no polynomial function $P \in \mathcal{P}_2(\mathbb{R})$ such that

   $$P(-2) = 0, \quad P(-1) = 0, \quad P(0) = 7, \quad , P(1) = 0, \quad P(2) = 0.$$

   (b) Find the least square approximation to this problem.

6. **Approximation by polynomials.**

   (a) Show that the following formula defines an inner product on $\mathcal{P}_3(\mathbb{R})$:

   $$\text{For } P, Q \in \mathcal{P}_3(\mathbb{R}), \quad \langle P, Q \rangle := \int_{-1}^{1} P(t)Q(t)\mathrm{d}t.$$

   (b) Find an orthogonal basis of the subspace $\mathcal{P}_2(\mathbb{R})$.

   (c) Compute the orthogonal projection of x^3 onto $\mathcal{P}_2(\mathbb{R})$.

7. **Orthogonal functions.** We work in the inner product space $\mathcal{C}_{2\pi}(\mathbb{R})$ introduced in Exercise 3.(b).

   (a) Recall that for all $x \in \mathbb{R}$, we have

   $$\cos(x) = \frac{e^{ix} + e^{-ix}}{2} \quad \text{and} \quad \sin(x) = \frac{e^{ix} - e^{-ix}}{2i}.$$

   For every $k, l \geq 1$, express the function $\sin(kx)\sin(lx)$ as a linear combination of functions of the form $\cos(nx)$ for $n \in \mathbb{Z}$.

   (b) Deduce that the family $(\sin(nx))_{n\geq 1}$ is orthogonal. What is the norm of the function $\sin(nx)$?

Remark: This last exercise provides the foundations for **Fourier Analysis**, the decomposition of periodic functions into series of simpler trigonometric functions.